

Mecklenburg County Sample



Traffic Impact Study

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EXECUTIVE SUMMARY

This Traffic Impact Analysis (TIA) evaluates the anticipated traffic impacts of the proposed Mecklenburg County Sample within Charlotte MSA, North Carolina (City of Charlotte / Mecklenburg County). The analysis follows the structure of the NCDOT Policy on Street and Driveway Access to North Carolina Highways (July 2003) Chapter 5 and the NCDOT Congestion Management Capacity Analysis Guidelines (Standards, March 2022; Best Practices, November 20, 2024). 9 study intersections within a 1.00-mile radius are analyzed under Existing, No-Build, and Build conditions.

9	1	3	6,800
INTERSECTIONS	LOS DROPS	AT LOS E/F	DAILY TRIPS (RAW)

1.0 INTRODUCTION

Project name	Mecklenburg County Sample
Land use	LU 820 — Shopping Center / Retail
Development size	85 1,000 sqft GLA
Site coordinates	35.1954°, -80.7643°
Build-out (horizon) year	2027
Controlling jurisdiction	City of Charlotte / Mecklenburg County
MPO	CRTPO (Charlotte Regional Transportation Planning Organization) — 2026-2035 TIP
Programmed projects source	NCDOT 2026-2035 STIP (adopted July 2025), as amended

1.1 TIA Applicability

Charlotte UDO Article 32, Section 32.1 (effective 6/1/2023) requires a Comprehensive Transportation Review (CTR) — Multimodal Assessment + TDM Assessment + Traffic Impact Study — when Charlotte Streets Manual Chapter 3, Table 3.1 thresholds are met (reported secondary-source values: >1,500 vpd or >150 peak-hour trips in lower-intensity districts; >2,000 vpd or >200 peak-hour trips in medium/high-intensity districts — confirm against the current Table 3.1 at scoping). The CTR applies at permitting even for by-right development.

The proposed development's estimated raw trip generation of 6,800 vehicles per day meets or exceeds the 3,000 vpd threshold at which NCDOT may require a TIA (no reductions are applied in the threshold test).

1.2 Scoping

CDOT publishes a Rezoning Transportation Analysis memo per rezoning petition; the TIS must be prepared by a North Carolina licensed Professional Engineer. NCDOT coordination applies on state-system streets. The background growth factor and the list of approved background developments must be agreed in the Approved Scoping Document prior to TIA submittal; a placeholder for that document is carried in the Appendix.

2.0 METHODOLOGY

NCDOT requires HCM-based analysis at submittal; the Department currently utilizes Synchro 11 (with mandatory SimTraffic simulation, minimum ten runs), HCS for freeway facilities, and Sidra for roundabouts. This screening applies the openly-published Webster/Akçelik signalized model pending those tool runs; Synchro 11-compatible models and SimTraffic outputs are to be provided at formal submittal.

Peak periods	AM and PM weekday peak hours (minimum); counts Tue–Thu, school in session, less than 12 months old
PHF (future conditions)	0.90 per Standards
Heavy vehicles	Average of duals and TTSTs, 2% minimum
RTOR (future analysis)	Not permitted per Standards
Turn-lane storage	Greater of Synchro 95th percentile and SimTraffic maximum queue, rounded up to 25 ft; 100 ft minimum
Roundabout threshold	v/c 0.85
Study-area rule	Intersections where site trips add 10% or more to background volumes on any approach/movement

2.1 Trip Generation Basis

NCDOT's Best Practices call for the ITE Trip Generation Manual with the Department's Rate-vs-Equation spreadsheet, internal capture per the NCHRP 684 spreadsheet method (vehicle occupancy 1.1, 4,000-ft maximum walking distance), and pass-by limited to retail uses with multi-use pass-by capped at 10% of the adjacent street volume. This screening analysis applies public-data trip-generation rates (NHTS 2017 / SANDAG 2002 / NCHRP 716), with every rate tagged so the jurisdiction-approved ITE figure can be substituted at submittal; applied internal-capture and pass-by credits must be reconciled against the NCDOT limits at scoping.

Daily rate applied	80.00 trips per ksf × 85 ksf
AM peak rate applied	3.20 trips per ksf × 85 ksf
PM peak rate applied	8.00 trips per ksf × 85 ksf
Rate source	SANDAG 2002 “(Not So) Brief Guide of Vehicular Traffic Generation Rates for the San Diego Region” — Community Shopping Center (daily 80/ksf; PM pass-by 30%)

3.0 ANALYSIS CONDITIONS AND MEASURES OF EFFECTIVENESS

Scenarios follow the NCDOT Standards set: Existing Base Year; No-Build Design Year (build-out); and Design Year Build. Where a scoped project is phased, intermediate build-out years are analyzed with earlier-phase trips carried as site trips (never background). Where an impacted STIP project is in planning, design, or construction (or within 5 years post-construction) and site traffic was not in the STIP forecast, an additional STIP design-year analysis is coordinated per the Best Practices TIA/STIP flow chart.

3.1 Intersection MOE Summary (NCDOT format)

Approaches are ordered EB, WB, NB, SB and movements left-through-right per the Best Practices reporting format. Signalized intersections report control delay and LOS overall and per lane group (any v/c > 1.0 is reported as LOS F regardless of delay); unsignalized intersections report no overall LOS — results are per lane group with the conflicting movement. SimTraffic maximum-queue columns are populated at submittal.

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
1	North Sharon Amity Road & Holbrook Drive	A / 4.9s	A / 4.9s	A / 5.3s
2	North Sharon Amity Road & Pierson Drive	A / 6.5s	A / 6.6s	A / 7.4s
3	North Sharon Amity Road & Charleston Drive	A / 7.8s	A / 7.9s	A / 8.2s
4	Albemarle Road & North Sharon Amity Road	F / 148.4s	F / 156.4s	F / 184.1s
5	Albemarle Road & Pierson Drive	A / 5.6s	A / 5.6s	A / 5.6s

No.	Intersection	Existing LOS/ Delay	No-Build LOS/ Delay	Build LOS/Delay
6	North Sharon Amity Road & Monroe Road	D / 51.0s	D / 54.4s	E / 65.2s
7	Idlewild Rd & Monroe Rd & Rama Rd	A / 7.5s	A / 7.6s	A / 7.8s
8	Idlewild Road & Connection Point Boulevard	A / 9.1s	A / 9.2s	A / 9.3s
9	Albemarle Rd & Reddman Rd	F / 300.0s	F / 300.0s	F / 300.0s

3.2 Trip Distribution — Gravity Model

The trip distribution uses the gravity model (term = $M / (d^{1.1} \cdot d_{site})$); mass basis: intersection through-volume (AADT × K-factor) as the destination-activity attraction proxy). Shares are normalized to 100% of project trips and drive the trip assignment below. Basis: NCHRP-716 gamma-friction gravity model (production-constrained) over study-area attraction zones.

Directional sector		Share of project trips			
Northeast (NNE + ENE)		60%			
Southeast (ESE + SSE)		21%			
Southwest (SSW + WSW)		17%			
Northwest (WNW + NNW)		2%			

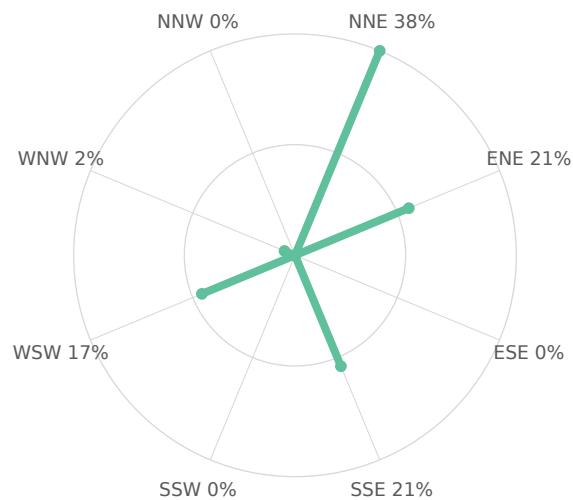
Study-area zone	Dir.	Distance (mi)	Mass (M)	Gravity term	Trip share
Albemarle Rd & Reddman Rd	ENE	0.98	4,365	0.2	21.36%
Albemarle Road & North Sharon Amity Road	NNE	0.47	3,510	0.2	20.79%
North Sharon Amity Road & Pierson Drive	NNE	0.14	2,610	0.2	17.59%
North Sharon Amity Road & Monroe Road	WSW	0.61	2,520	0.1	14.32%
Idlewild Rd & Monroe Rd & Rama Rd	SSE	0.80	2,025	0.1	10.86%
Idlewild Road & Connection Point Boulevard	SSE	0.85	1,890	0.1	9.96%
North Sharon Amity Road & Charleston Drive	WSW	0.19	477	0.0	3.16%
North Sharon Amity Road & Holbrook Drive	WNW	0.02	198	0.0	1.39%
Albemarle Road & Pierson Drive	WNW	0.58	100	0.0	0.58%

9 study-area zone(s). Screening-grade distribution; not a calibrated regional model.



Figure — Project Trip Distribution. Study-area zones plotted to scale at their true bearing and distance from the site; leg weight is proportional to each zone's share of project trips, and the label gives that share. Derived from the Gravity Model distribution — the same shares tabulated above. Screening-grade: zone positions are the analysis locations, not a surveyed base map.

Figure — Directional Distribution of Project Trips

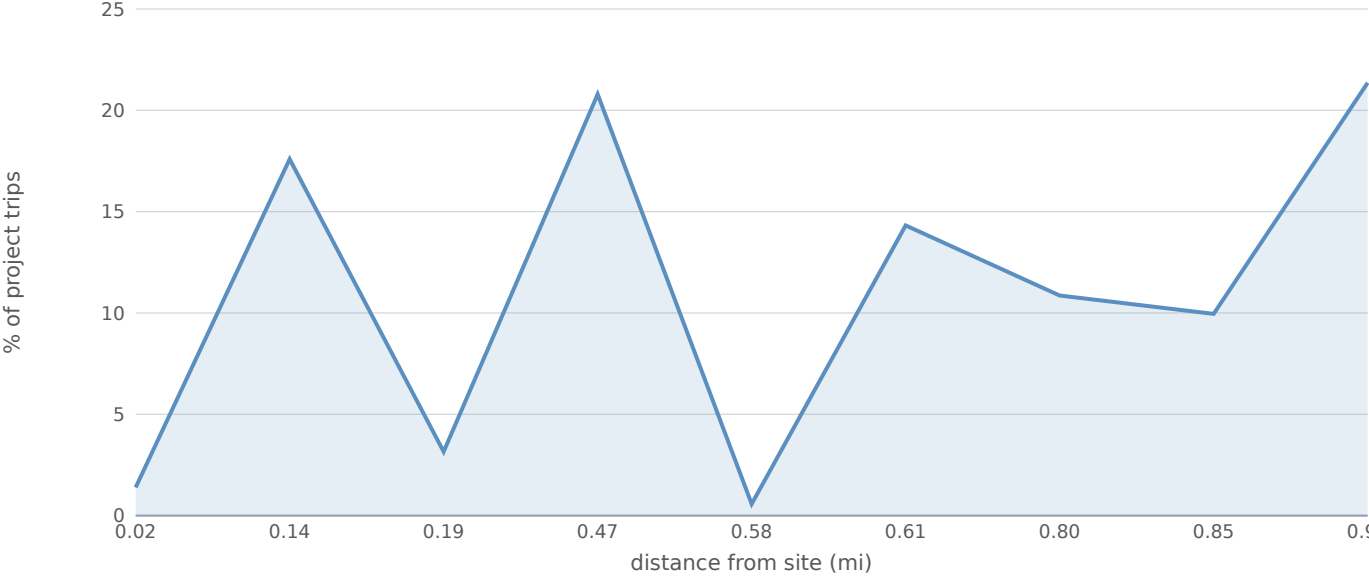


Screening-grade directional distribution of net new project trips by compass octant (spoke length $\propto$ percent of project trips). Derived from the Gravity Model distribution.

Figure — Project Trip Share by Study-Area Zone



Figure — Trip Share vs. Distance from Site (Gravity Decay)



3.3 Project Trip Assignment

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
North Sharon Amity Road & Holbrook Drive	WNW	0.02	129	193	26.0%
North Sharon Amity Road & Pierson Drive	NNE	0.14	111	199	26.9%
North Sharon Amity Road & Charleston Drive	WSW	0.19	32	80	10.8%
Albemarle Road & North Sharon Amity Road	NNE	0.47	78	116	15.7%
Albemarle Road & Pierson Drive	WNW	0.58	0	1	0.1%

Study intersection	Dir.	Distance (mi)	AM trips	PM trips	Share of project
North Sharon Amity Road & Monroe Road	WSW	0.61	50	74	10.0%
Idlewild Rd & Monroe Rd & Rama Rd	SSE	0.80	17	44	5.9%
Idlewild Road & Connection Point Boulevard	SSE	0.85	8	12	1.6%
Albemarle Rd & Reddman Rd	ENE	0.98	12	22	3.0%

Project trips assigned to each study intersection from the distribution shares above: a four-step gravity model sets the share reaching each intersection from its attraction mass and its distance from the site. Directional load multipliers are a Florida (Caltran) refinement and are not applied in this region — each intersection's loading is taken directly from the gravity weights above.

4.0 MITIGATION CRITERIA CHECK (POLICY CH. 5.J)

NCDOT requires improvements when, comparing base-network and project conditions: total average delay increases by 25% or more while remaining at the same LOS; the LOS degrades by one level; or the intersection operates at LOS F. Turn-lane mitigation applies where the 95th-percentile queue exceeds existing storage. Signal-timing changes alone are not considered mitigation. The District Engineer makes the final mitigation determination.

No.	Intersection	Δ delay	LOS drop	LOS F	Ch. 5.J result
1	North Sharon Amity Road & Holbrook Drive	+8%	no	no	none indicated
2	North Sharon Amity Road & Pierson Drive	+12%	no	no	none indicated
3	North Sharon Amity Road & Charleston Drive	+4%	no	no	none indicated
4	Albemarle Road & North Sharon Amity Road	+18%	no	YES	MITIGATION INDICATED
5	Albemarle Road & Pierson Drive	+0%	no	no	none indicated
6	North Sharon Amity Road & Monroe Road	+20%	YES (D→E)	no	MITIGATION INDICATED
7	Idlewild Rd & Monroe Rd & Rama Rd	+3%	no	no	none indicated
8	Idlewild Road & Connection Point Boulevard	+1%	no	no	none indicated
9	Albemarle Rd & Reddman Rd	+0%	no	YES	MITIGATION INDICATED

5.0 CONCLUSIONS AND SUBMITTAL NOTES

1 intersection project a LOS drop and 3 operate at LOS E or F under Build conditions; mitigation per Policy Ch. 5.J should be developed with the District Engineer.

Data conventions: NCDOT forecasts supply AADT, % duals and TTSTs, directional split (D), and design-hour factor (K); forecast conversion to peak-hour volumes uses the Intersection Analysis Utility (IAU) in the NCDOT Traffic Engineering Suite. AADT reference: NCDOT ArcGIS AADT mapping application. Crash analysis: a site-radius crash assessment from NCDOT's TEAAS (Traffic Engineering Accident Analysis System) is prepared at submittal — not generated by this screening tool.

This TIA is to be prepared under the direct charge of and sealed by a licensed North Carolina Professional Engineer with expertise in traffic engineering (Policy Ch. 5.A; 21 NCAC 56 .1103 — seal with signature over or adjacent to the seal and date of signing; firm name, address, and NC firm license number on the title block). TIAs marked confidential remain so until a driveway permit is requested or the project is publicly announced.

SCREENING DISCLOSURE: this document is a screening-level draft. It does not include the Approved Scoping Document, SimTraffic outputs, Synchro models, TEAAS crash analysis, or signed/sealed certification — all required for a formal NCDOT TIA submittal.

FOUR-STEP TRAVEL DEMAND MODEL

Off-site project trips are distributed and assigned with the four-step urban transportation modeling process (FHWA; NCHRP Report 716, Travel Demand Forecasting). Each step below shows what the engine computed for this site.

Step 1 — Trip Generation

Public-data screening trip rates (SANDAG 2002, corroborated by NHTS 2017 / NCHRP 716) applied to the proposed land use give the site's productions.

Land use	LU 820 — Shopping Center / Retail
Size	85 1,000 sqft GLA
Daily trips (productions)	6,800
PM peak-hour trips	680 (326 in / 354 out)

Step 2 — Trip Distribution (Gravity Model)

A production-constrained gravity model allocates trips to surrounding signals: $T_j = P \cdot (A_j \cdot F_j) / \sum (A_x \cdot F_x)$. Attractiveness A_j is each signal's through-volume; the friction factor F_j is a gamma function $F = a \cdot t^b \cdot e^{(c \cdot t)}$ — the form recommended by NCHRP Report 716 — on the travel time t to the signal, with home-based-work shape parameters $b=-0.02$, $c=-0.123$ (consistent with NCHRP 716's sampled MPO gravity parameters; the scale constant a cancels in the share normalization).

Signal (attraction zone)	Dist mi	Time min	PM trips	Share
North Sharon Amity Road & Pierson Drive	0.14	0.3	199	26.9%
North Sharon Amity Road & Holbrook Drive	0.02	0.1	193	26.0%
Albemarle Road & North Sharon Amity Road	0.47	1.1	116	15.7%
North Sharon Amity Road & Charleston Drive	0.19	0.5	80	10.8%
North Sharon Amity Road & Monroe Road	0.61	1.5	74	10.0%
Idlewild Rd & Monroe Rd & Rama Rd	0.80	1.9	44	5.9%
Albemarle Rd & Reddman Rd	0.98	2.4	22	3.0%
Idlewild Road & Connection Point Boulevard	0.85	2.0	12	1.6%
Albemarle Road & Pierson Drive	0.58	1.4	1	0.1%

Top 12 distribution zones by assigned PM-peak trips shown; the full set is in the capacity appendix.

Step 3 — Mode Choice

Auto-mode share 85% applied via a binary logit $P(\text{auto}) = 1 / (1 + e^{-(\text{ASC} - \lambda \cdot \Delta \text{GC})})$ calibrated to the metro's auto-mode share — measured (ACS B08301) where the metro is wired, otherwise the ACS B08301 metropolitan median of 90% — and shifted by site urbanity (local density), so a denser, more transit-served site splits further from auto than a greenfield parcel in the same metro. The remaining 15% of trips (transit, walking, cycling) do not load the off-site roadway network; only auto trips are carried into Step 4.

Step 4 — Route Assignment

Project trips are routed over the local road network by shortest path and loaded with a capacity-constrained BPR equilibrium (volume-delay $t = t_0 \cdot [1 + 0.15 \cdot (v/c)^4]$, Method of Successive Averages), so congested links shift trips toward alternative routes. 88.9% of study signals lie on a modeled route; highest loaded-link v/c : 0.36. Corridors carrying the project trips:

Corridor (road class)	Loaded length mi	PM project vph	v/c
Minor arterial	2.84	401	0.36

Corridor (road class)	Loaded length mi	PM project vph	v/c
Collector	0.01	231	0.27
Trunk highway	0.62	86	0.21

Network shortest-path assignment over the OSM road graph within the study area; per-signal v/c, delay, LOS, and queue are in the capacity appendix.

Conserved assignment: project trips routed to 15 cordon gateways on the study boundary (weighted by the directional distribution above). Flow conservation verified at render input: Σ entering = Σ leaving at 470 network junctions, max imbalance 0.00 veh. 9 study intersection(s) carry path-derived movements; 0 retain the geometric octant model (labeled per intersection).

APPENDIX — INTERSECTION CAPACITY ANALYSIS WORKSHEETS

One worksheet per study intersection: a turning-movement diagram for each analyzed peak period plus the per-approach capacity table. Analysis uses the openly-published Webster/Akcelik signalized control-delay method; v/c , control delay (sec/veh), Level of Service, and 95th-percentile back-of-queue (ft) are reported for the Existing (No-Build) and Build conditions.

Study scope: 9 signalized intersections lie within the 1-mile study area, and all 9 are carried as study intersections. The regional signal inventory holds 27 records inside that radius; 18 were merged into a junction already counted, because a signalized junction is commonly recorded once per approach rather than once per intersection. Merged records are duplicates of an analyzed intersection, not intersections omitted from the study.

Background approach volumes are resolved per leg — the signal's design hour on the main road (its counted volume where the analyzer had a compatible count, else the road-class baseline it assigned — each worksheet says which), the road-class baseline on uncounted legs, client link counts where supplied; half per direction, and half in the physical direction of a one-way carriageway — and the background turning movements in the diagrams are balanced to the exit legs by iterative proportional fitting (NCHRP 255/765 refinement) from a geometry seed; each worksheet states its leg sources and residual. Project-trip movements are assigned geometrically from the study's directional trip distribution (see each worksheet's Affected movements table). Replace both with measured turning-movement counts (TMCs) before a formal submittal.

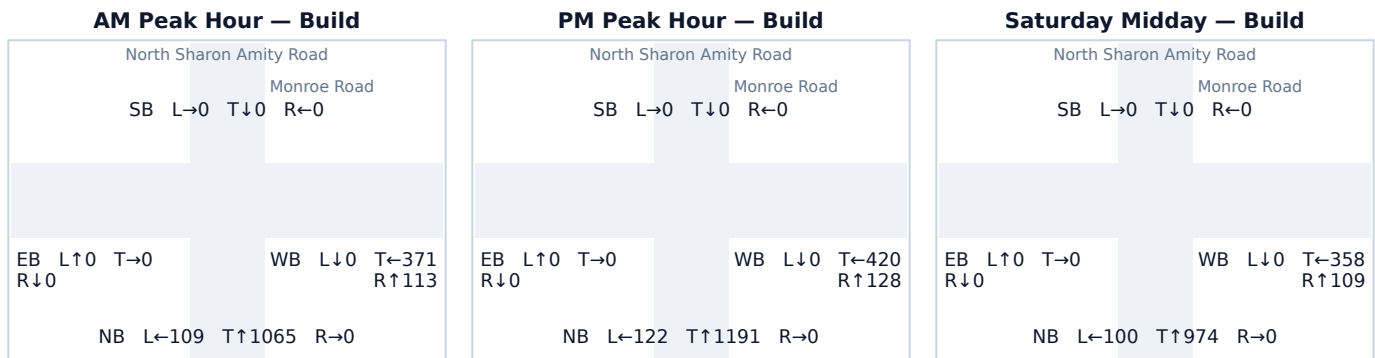
A.1 North Sharon Amity Road & Monroe Road

Signal ID	charlotte-5527699422
Location	NPA 52 · 0.61 mi from site
Added PM peak trips	74
Existing / No-Build (PM)	LOS D · 54.4 s/veh · v/c 0.99
Build (PM)	LOS E · 65.2 s/veh · v/c 1.03
Δ control delay	+10.8 s/veh (LOS change)
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: design year (opening year +10.8s, design year +17.9s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 0.99 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 8 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	1,143	117	0	1,260
SB approach	0	—	0	0	0
EB approach	0	0	—	0	0
WB approach	0	117	383	—	500
Σ exiting	0	1,260	500	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	1,279	34	1,313	1.58 → 1.62	291.0 → 300.0	F → F	1,343
SB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
EB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
WB	508	40	548	0.63 → 0.68	22.6 → 24.1	C → C	447

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Right	40	54%
NB Through	34	46%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

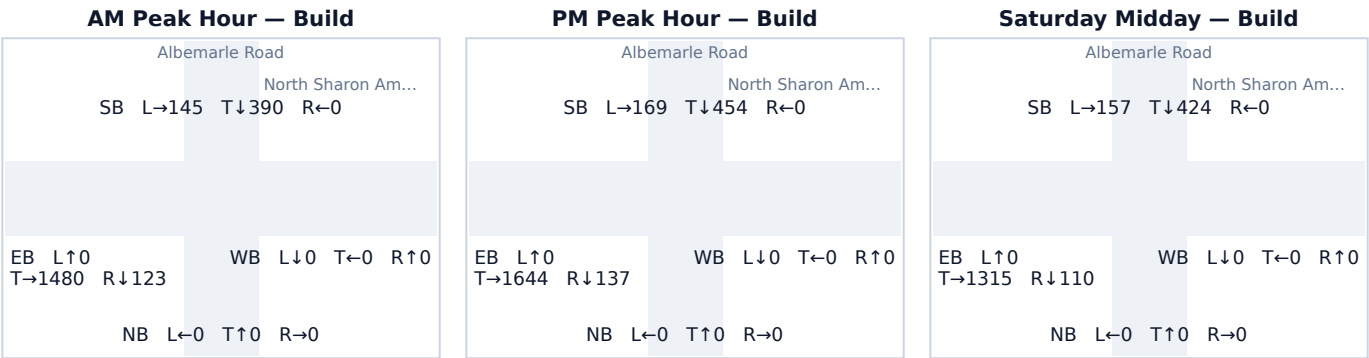
A.2 Albemarle Road & North Sharon Amity Road

Signal ID	charlotte-2181076171
Location	NPA 16 · 0.47 mi from site
Added PM peak trips	116
Existing / No-Build (PM)	LOS F · 156.4 s/veh · v/c 1.27
Build (PM)	LOS F · 184.1 s/veh · v/c 1.34
Δ control delay	+27.7 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +27.7s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.27 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 8 iterations, residual 0.2 vph). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	0	0	0
SB approach	365	—	0	136	500
EB approach	136	0	—	1,619	1,755
WB approach	0	0	0	—	0
Σ exiting	500	0	0	1,755	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0
SB	508	116	623	0.63 → 0.77	22.6 → 27.8	C → C	541
EB	1,781	0	1,781	2.20 → 2.20	300.0 → 300.0	F → F	1,822
WB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
SB Through	116	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

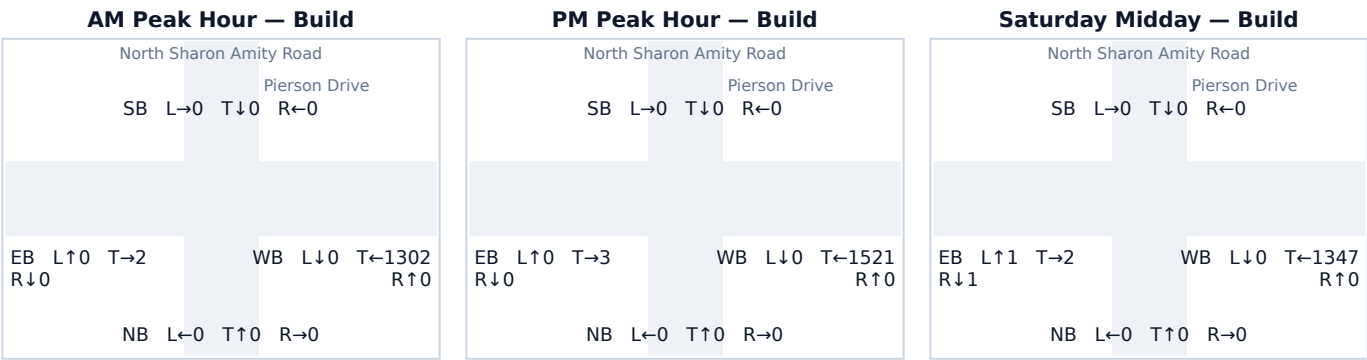
A.3 North Sharon Amity Road & Pierson Drive

Signal ID	charlotte-2181076118
Location	NPA 16 · 0.14 mi from site
Added PM peak trips	199
Existing / No-Build (PM)	LOS A · 6.6 s/veh · v/c 0.47
Build (PM)	LOS A · 7.4 s/veh · v/c 0.54
Δ control delay	+0.8 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.8s, design year +1.5s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 80 s cycle, 2 critical phases; g/C NS 0.17 / EW 0.70; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 2 from the signal's counted design hour (half per direction). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph). 2 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ EB leg	→ WB leg	Σ entering
EB approach	—	0	0
WB approach	1,305	—	1,305
Σ exiting	1,305	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	27.3 → 27.3	C → C	0
SB	0	0	0	0.00 → 0.00	27.3 → 27.3	C → C	0
EB	0	3	3	0.00 → 0.00	3.6 → 3.6	A → A	1
WB	1,325	196	1,521	1.05 → 1.20	50.7 → 111.1	D → F	1,362

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	196	98%
EB Left	3	2%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

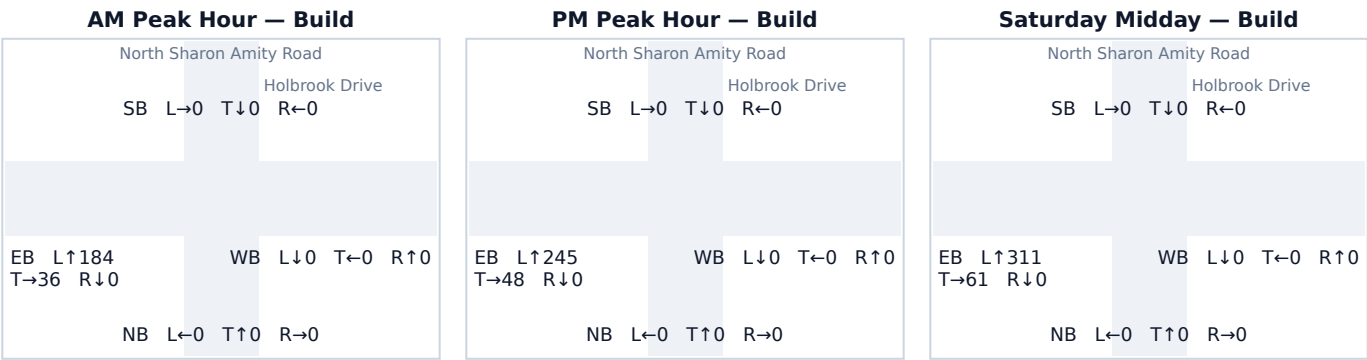
A.4 North Sharon Amity Road & Holbrook Drive

Signal ID	charlotte-172660814
Location	NPA 16 · 0.02 mi from site
Added PM peak trips	193
Existing / No-Build (PM)	LOS A · 4.9 s/veh · v/c 0.04
Build (PM)	LOS A · 5.3 s/veh · v/c 0.12
Δ control delay	+0.4 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.4s, design year +0.4s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.60; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph; exits scaled to entries, 505% imbalance). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ SB leg	→ EB leg	→ WB leg	Σ entering
SB approach	—	0	0	0
EB approach	83	—	16	99
WB approach	0	0	—	0
Σ exiting	83	0	16	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
SB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
EB	101	193	293	0.09 → 0.27	5.2 → 6.3	A → A	96
WB	0	0	0	0.00 → 0.00	4.7 → 4.7	A → A	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	193	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

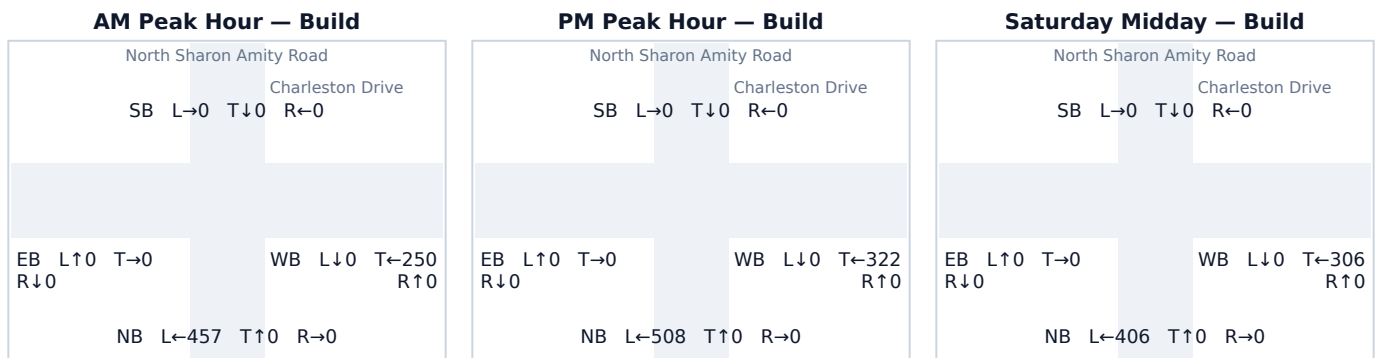
A.5 North Sharon Amity Road & Charleston Drive

Signal ID	charlotte-7810838990
Location	NPA 52 · 0.19 mi from site
Added PM peak trips	80
Existing / No-Build (PM)	LOS A · 7.9 s/veh · v/c 0.33
Build (PM)	LOS A · 8.2 s/veh · v/c 0.37
Δ control delay	+0.3 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.3s, design year +0.4s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.56 / EW 0.27; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph; exits scaled to entries, 68% imbalance). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	500	0	500
EB approach	0	—	0	0
WB approach	0	239	—	239
Σ exiting	0	739	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	508	0	508	0.50 → 0.50	9.7 → 9.7	A → A	212
SB	0	0	0	0.00 → 0.00	5.7 → 5.7	A → A	0
EB	0	0	0	0.00 → 0.00	16.0 → 16.0	B → B	0
WB	242	80	322	0.50 → 0.67	22.2 → 26.6	C → C	197

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	80	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

A.6 Idlewild Rd & Monroe Rd & Rama Rd

Signal ID	charlotte-7945928189
Location	NPA 52 · 0.80 mi from site
Added PM peak trips	44
Existing / No-Build (PM)	LOS A · 7.6 s/veh · v/c 0.43
Build (PM)	LOS A · 7.8 s/veh · v/c 0.44
Δ control delay	+0.2 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.2s, design year +0.3s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.60; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 2 from the signal's counted design hour (half per direction). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph). 2 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ EB leg	→ WB leg	Σ entering
EB approach	—	1,013	1,013
WB approach	0	—	0
Σ exiting	0	1,013	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
SB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
EB	1,028	44	1,071	0.95 → 0.99	28.4 → 36.3	C → D	724
WB	0	0	0	0.00 → 0.00	4.7 → 4.7	A → A	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Through	44	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

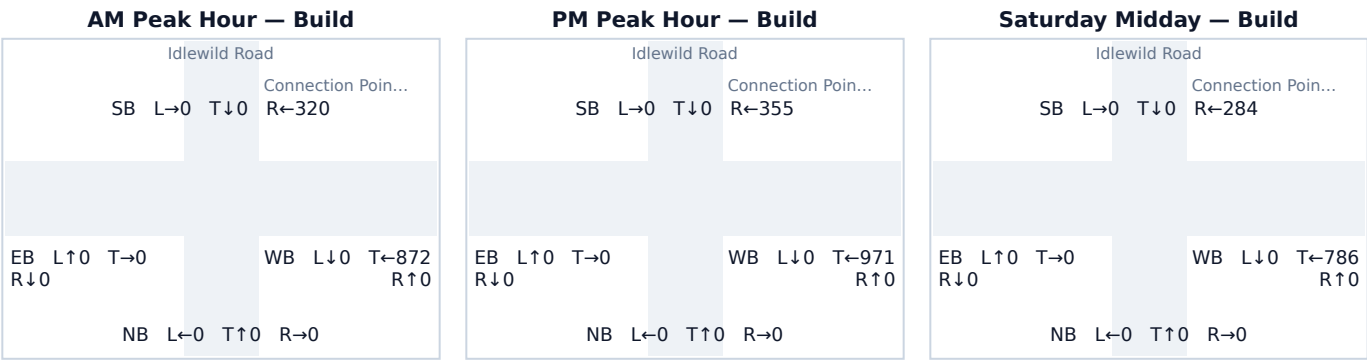
A.7 Idlewild Road & Connection Point Boulevard

Signal ID	charlotte-5516526369
Location	NPA 52 · 0.85 mi from site
Added PM peak trips	12
Existing / No-Build (PM)	LOS A · 9.2 s/veh · v/c 0.52
Build (PM)	LOS A · 9.3 s/veh · v/c 0.52
Δ control delay	+0.1 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.1s, design year +0.1s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 75 s cycle, 2 critical phases; g/C NS 0.23 / EW 0.63; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: balanced estimate (Furness/IPF, 1 iterations, residual 0.0 vph; exits scaled to entries, 27% imbalance). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ SB leg	→ EB leg	→ WB leg	Σ entering
SB approach	—	350	0	350
EB approach	0	—	0	0
WB approach	0	945	—	945
Σ exiting	0	1,295	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	22.0 → 22.0	C → C	0
SB	355	0	355	0.84 → 0.84	45.6 → 45.6	D → D	291
EB	0	0	0	0.00 → 0.00	5.1 → 5.1	A → A	0
WB	959	12	971	0.84 → 0.85	18.5 → 19.2	B → B	667

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	12	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

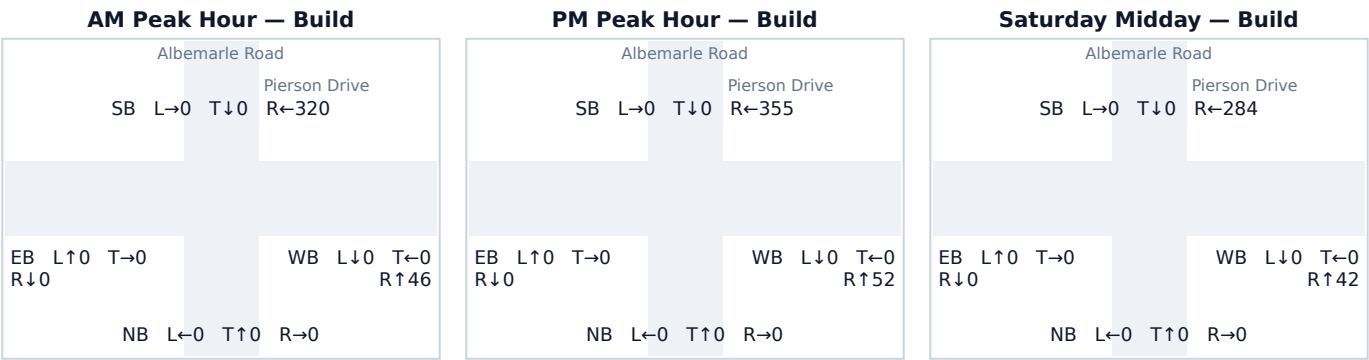
A.8 Albemarle Road & Pierson Drive

Signal ID	charlotte-1711278797
Location	NPA 240 · 0.58 mi from site
Added PM peak trips	1
Existing / No-Build (PM)	LOS A · 5.6 s/veh · v/c 0.17
Build (PM)	LOS A · 5.6 s/veh · v/c 0.17
Δ control delay	+0.0 s/veh
Mitigation	No mitigation indicated at screening level (opening year +0.0s, design year +0.0s). The largest projected delay change is below the 5-second screening threshold — a screening default, not an agency criterion: no jurisdiction was resolved for this site. The governing agency's TIS criteria supersede this, and typically add a v/c test and a separate clause for approaches already failing in the no-build; neither is screened here.

Signal timing: a Webster optimum cycle with Critical Movement Method splits derived from the no-build approach volumes (FHWA-HOP-07-006) — 60 s cycle, 2 critical phases; g/C NS 0.60 / EW 0.23; permissive NS / permissive EW left-turn phasing (inferred from the FHWA-HRT-04-091 cross product of left-turn and opposing through volume); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 3 from the signal's counted design hour (half per direction); 1 of 3 from the road-class baseline (no count on that leg). Turning movements: did not balance within 50 iterations (residual 300.0 vph; rows held exact). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured period turning-movement counts.

Balanced turning movements (vph)	→ SB leg	→ EB leg	→ WB leg	Σ entering
SB approach	—	350	0	350
EB approach	0	—	0	0
WB approach	50	0	—	50
Σ exiting	50	350	0	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	0	0	0	0.00 → 0.00	4.7 → 4.7	A → A	0
SB	355	0	355	0.33 → 0.33	6.7 → 6.7	A → A	121
EB	0	0	0	0.00 → 0.00	17.7 → 17.7	B → B	0
WB	51	1	52	0.12 → 0.12	18.9 → 18.9	B → B	28

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
WB Through	1	100%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.

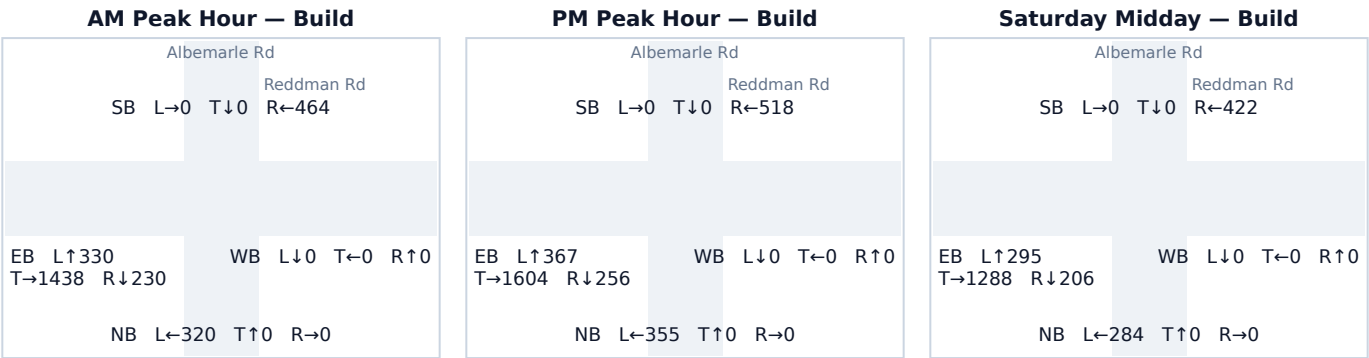
A.9 Albemarle Rd & Reddman Rd

Signal ID	charlotte-172211825
Location	NPA 164 · 0.98 mi from site
Added PM peak trips	22
Existing / No-Build (PM)	LOS F · 300.0 s/veh · v/c 1.71
Build (PM)	LOS F · 300.0 s/veh · v/c 1.72
Δ control delay	+0.0 s/veh
Mitigation	Major: add a dedicated turn lane on the critical approach AND retime the signal; reconsider site driveway alignment or development scale if delay remains above 80s. Governing horizon: opening year (opening year +0.0s, design year +0.0s).

Signal timing: the generic 90 s / g/C 0.45 screening default — critical flow ratio 1.71 puts this intersection at or beyond saturation, where Webster is not applicable — 90 s cycle, 2 critical phases; g/C NS 0.45 / EW 0.45; permissive NS / permissive EW left-turn phasing (screening default); pedestrian minimum green 14 s NS / 14 s EW. Timing is resolved once from the no-build volumes and held fixed across scenarios; approach capacity is 1,800 pc/h/ln × that phase's g/C × through lanes × weather factor (one through lane per approach — no lane count was available).

Leg volumes: 2 of 4 from the signal's counted design hour (half per direction); 2 of 4 from the road-class baseline (no count on that leg). Turning movements: did not balance within 50 iterations (residual 419.1 vph; rows held exact); exits scaled to entries, 72% imbalance. 1 extra leg not carried (4×4 matrix). A one-way carriageway carries half of the two-way count in its direction (a one-way couplet street is understated).

Turning-Movement Diagrams (Build condition)



Background turning-movement volumes vary by period: the stored design-hour count (AADT × K-factor, ≈ the PM peak) is carried at 100% for the PM peak and at screening-level shares of the design hour for the AM peak (90%) and Saturday midday (80%), reflecting that the network is not equally loaded across the day. A submitted study replaces these with measured per-period turning-movement counts.

Balanced turning movements (vph)	→ NB leg	→ SB leg	→ EB leg	→ WB leg	Σ entering
NB approach	—	0	350	0	350
SB approach	0	—	500	0	500
EB approach	252	360	—	1,571	2,183
WB approach	0	0	0	—	0
Σ exiting	252	360	850	1,571	

Design-hour basis — existing year, before growth and period scaling, so it is identical for every analysis period and will not tie to the grown No-Build volumes above. Columns are the leg each movement exits through; the diagonal (U-turn) is folded into the left turn. Entries equal exits at this junction within the stated residual.

Per-Approach Capacity (PM Peak)

Appr.	NB vph	+Trips	Build vph	v/c NB→Bld	Delay NB→Bld	LOS NB→Bld	Q95 ft
NB	355	0	355	0.44 → 0.44	18.7 → 18.7	B → B	251
SB	508	11	518	0.63 → 0.64	22.6 → 23.0	C → C	413
EB	2,215	11	2,227	2.73 → 2.75	300.0 → 300.0	F → F	2,278
WB	0	0	0	0.00 → 0.00	13.6 → 13.6	B → B	0

NB = No-Build: existing counts grown to the opening year, without project trips. Build = No-Build plus assigned project trips. Current-year Existing conditions appear in the scenario tables of the main report body.

Affected movements (PM peak project trips)

Movement	Project trips	% of project trips
EB Left	11	50%
SB Right	11	50%

Source: routed network paths through this junction (conserved assignment — the trips in these rows are the same vehicles counted at the adjacent studied junctions along their paths).

Movement loads are derived from the study's directional trip distribution and the site's bearing from this intersection (outbound trips enter from the site leg and turn toward their destination sector; inbound trips mirror). U-turns are folded into the left-turn movement. The same assignment drives the per-approach capacity loading above, so the movement rows cross-foot with the junction's added project trips in total and with the +Trips column approach-by-approach (each trip is counted on its entering approach). The turning-movement diagrams show TOTAL approach volumes under the screening 15/70/15 split, not the project increment. Screening-level: replace with measured turning-movement counts and the site's access-point routing at submittal.